

Program : Diploma in Cloud Computing and Big Data	
Course Code : 4279	Course Title: Advanced Operating System
Semester : 4	Credits: 0
Course Category: Programme Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Provide an exposure to the basics of Virtualization Technology and Cloud Computing.
- Provide knowledge to apply security and storage infrastructure in Cloud Computing Platforms
- Provide an exposure to the fundamentals of dockers and containers

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basics of Operating Systems		Operating System	3
Linux Administration		Linux System Administration Lab	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basic concepts of Virtualization and Hypervisors	13	Understanding
CO2	Summarize Virtual Machines and Network, Desktop and Application Virtualization	15	Understanding
CO3	Describe the storage infrastructure and management activities	15	Understanding

CO4	Extend the concepts of Docker, Container, Kubernetes	15	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basic concepts of Virtualization and Hypervisors		
M1.01	Explain Virtualization and its importance	3	Understanding
M1.02	Outline Virtualization and Cloud Computing	2	Understanding
M1.03	Summarize Virtualization Software Operation and Types of Virtualization	2	Understanding
M1.04	Classify Hypervisors	2	Understanding
M1.05	Describe the role of a Hypervisor	2	Understanding
M1.06	Compare Hypervisors	2	Understanding
Contents: Understanding Virtualization – Describing Virtualization – Importance of Virtualization – Virtualization and Cloud Computing – Virtualization Software Operation – Types of Virtualization Understanding Hypervisors – Describing Hypervisor – Type 1 and Type 2 Hypervisors – Role of a Hypervisor – Comparison of Hypervisors			

CO2	Summarize Virtual Machines and Network, Desktop and Application Virtualization		
M2.01	Summarize Virtual Machines	2	Understanding
M2.02	Outline how to examine CPUs, Memory, Network and Storage in a VM.	1	Understanding
M2.03	Explain VM Clones, Templates and Snapshots	1	Understanding
M2.04	Describe Open Virtualization Format	1	Understanding
M2.05	Summarize the benefits and components of Network Virtualization	1	Understanding
M2.06	Outline Virtual Switch, Virtual LAN and VM Migration Services	2	Understanding
M2.07	Explain Advantages and Limitations of Desktop Virtualization	2	Understanding
M2.08	Outline Techniques used for Desktop Virtualization (RDS and VDI)	1	Understanding
M2.09	Summarize Components for Desktop Virtualization	1	Understanding
M2.10	Explain Advantages and Limitations of Application Virtualization and Tools used.	2	Understanding
	Series Test – I	1	
Contents : Understanding Virtual Machines (VM) – Describing VM – CPUs, Memory, Network and Storage in a VM – Working of VM – VM Clones, Templates, Snapshots – Open Virtualization Format Network Virtualization – Benefits – Components – Virtual Switch – Virtual LAN – VM Migration Services Desktop and Application Virtualization – Advantages and Limitations of Desktop Virtualization – Techniques used for Desktop Virtualization (RDS and VDI) – Components for Desktop Virtualization – Advantages and Limitations of Application Virtualization – Tools used for Application Virtualization			
CO3	Describe the storage infrastructure and management activities		
M3.01	Explain Security Framework	2	Understanding
M3.02	Define Risk Triad	2	Remembering
M3.03	Outline Storage Security Domains	3	Understanding
M3.04	Describe Security Implementations in Storage Networking	6	Understanding

M3.05	Summarize security mechanisms in Virtualized and Cloud Environments	2	Understanding
Contents : Securing and Storage Infrastructure: Information Security Framework, Risk Triad, Storage Security Domains, Security Implementations in Storage Networking, Securing Storage Infrastructure in Virtualized and Cloud Environments.			
CO4	Extend the concepts of Docker, Container, Kubernetes		
M4.01	Outline fundamentals of Docker	2	Understanding
M4.02	Relate Docker Commands	3	Understanding
M4.03	Define Docker Engine	3	Remembering
M4.04	Describe introduction about Container	3	Understanding
M4.04	Explain Kubernetes Architecture	3	Understanding
	Series Test – II	1	
Contents : Introduction to Docker, Docker Commands, Docker Engine, Container, Kubernetes Architecture			

Text / Reference

T/R	Book Title/Author
T1	Matthew Portnoy, <i>Virtualization Essentials</i> , 2 nd Edition, Sybex (Wiley) Publication, 2016
T2	Shailendra Singh, <i>Cloud Computing</i> , Oxford University Press, 2018
T3	Information Storage and Management, Author : EMC Education Services, Publisher: Wiley ISBN: 9781118094839
T4	Rajkumar Buyya, Christian Vecchiola and ThamariSelvi S, —Mastering in Cloud Computing, McGraw Hill Education (India) Private Limited, 2013
R1	Docker Easy: The Complete Guide on Docker World for Beginners
R2	Dan C. Marinescu, <i>Cloud Computing: Theory and Practice</i> , Elsevier Science, 2013, 1st Edition, Print Book ISBN :9780124046276, eBook ISBN :9780124046412

Online Resources

Sl.No	Website Link
1	https://www.tutorialspoint.com/virtualization2.0/index.htm
2	http://lsi.vc.ehu.es/pablogn/docencia/manuales/The%20Docker%20Book.pdf